



REVIEW ARTICLE

Approaches to forecast soil nutrient dynamics for precision agriculture and sustainable fertiliser management: A review

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Abstract

Predictive modelling of soil nutrient dynamics is an essential tool for promoting sustainable agricultural practices and environmentally responsible farming methods. The statistical and machine learning techniques used to forecast the availability and dynamics of soil nutrients are summarised in this review. The core frameworks for measuring spatio-temporal nutritional variability are established by traditional statistical approaches such as time-series models autoregressive integrated moving average (ARIMA), seasonal autoregressive integrated moving average (SARIMA), multivariate techniques (Principal component analysis (PCA) and factor analysis) and geostatistical tools (kriging). By capturing intricate nonlinear interactions within heterogeneous agroecosystems, machine learning techniques like random forest, support vector machines and ensemble approaches (XGBoost, LightGBM and AdaBoost) provide higher prediction accuracy. Forecasting capabilities are further enhanced by hybrid frameworks [Autoregressive integrated moving average with exogenous variables–artificial neural network (ARIMAX-ANN)] and deep learning architectures (Convolutional neural network (CNN), long short-term memory (LSTM), ANN). With R^2 values above 0.93 and notable decreases in prediction errors, ensemble approaches routinely perform better than traditional linear models. Nevertheless, persistent challenges include data quality limitations, spatial sampling constraints, insufficient environmental covariates and reduced model transferability across diverse pedoclimatic regions. Integrating high-resolution soil properties, climatic variables, terrain attributes and spectral information with advanced modelling architectures remains crucial for enhancing predictive reliability, ultimately supporting precision nutrient management, improved fertiliser efficiency and environmentally responsible agricultural systems.

Keywords: precision agriculture; predictive modelling; soil fertility; soil nutrient dynamics; sustainable nutrient management

Introduction

Sustainable land management in agricultural areas depends critically on maintaining healthy soils, which is essential for effective crop production. Healthy soils minimise dependence on external inputs, lower nutrient losses to the environment and increase workable days for agricultural activities (1). A healthy soil ecosystem underpins ecosystem stability and resilience, thereby supporting biological productivity and providing vital ecosystem services (2). Today, soils are seen as functional ecosystems that support important carbon, water and nutrient cycles, giving rise to the concept of soil health (3). As our understanding of soil science advanced, soil testing expanded beyond macronutrients [nitrogen (N), phosphorus (P) and (K)] to include secondary nutrients, micronutrients, pH, soluble salts and soil organic matter to better determine nutrient availability (4). A proper evaluation of nutrient availability in soils is important to optimise fertiliser use efficiency while ensuring environmental protection. Considering global

environmental changes, which affect nutrient cycles, quantifying nutrient availability is crucial to deepen our understanding of biogeochemical cycles across both natural and managed agroecosystems (5). Soil nutrient dynamics encompass the processes controlling the transformation, translocation, storage and bioavailability of soil nutrients, which fundamentally dictate plant nutrition, soil fertility and environmental quality (6). These dynamics govern how soil nutrients are stored, transported and absorbed by plant roots (7). Soil nutrients play an essential role in soil fertility and serve as reliable indicators of overall soil health (8). Soil N, P and K contents are pivotal drivers of crop growth, mediating both natural nutrient cycling and responses to fertiliser applications (9). Understanding soil nutrient pool dynamics is essential to prevent nutrient depletion or toxic accumulation, balancing plant uptake with fertiliser applications. These insights are vital for advancing sustainable agriculture, given that soil nutrient availability directly governs crop productivity (10). Consequently, there is an increasing need to develop quick, precise and cost-effective methodologies to

assess and forecast soil nutrient status, particularly given the significance of these dynamics in optimising fertiliser use, improving crop yields and mitigating environmental impacts (11).

Driven by advancements in data acquisition technologies and computational methods, predictive modelling of soil nutrient dynamics has become an essential tool in the decision-making process in agriculture today. Predictive modelling is the application of statistical and machine learning methods for estimating the levels of nutrients in the soil and their dynamics based on data from various sources, including remote sensing, spectroscopy, environmental covariates and *in situ* sensors (12). Current predictive modelling frameworks for soil nutrient dynamics incorporate various statistical and machine learning methods, such as linear regression, generalised additive model (GAM), partial least squares regression (PLSR), random forest (RF), support vector machines (SVM), boosting models and time series models, among others, in the estimation of the complex changes in the nutrient levels in the soil. In diverse agro-ecosystems, ensemble and non-linear models such as RF extreme gradient boosting (XGBoost), adaptive boosting (AdaBoost) and Hybrid statistical-machine learning models have demonstrated superior performance in comparison to conventional linear models. The superior performance of these models is reflected in enhanced predictive accuracy, the capacity to handle non-linear relationships and superior characterisation of nutrient heterogeneity (13). This underscores that integrating advanced statistical models with soil properties, climatic variables, terrain attributes and remote sensing imagery provides a robust foundation for soil nutrient forecasting. Consequently, this approach facilitates the design of effective nutrient management strategies to advance sustainable agricultural practices.

Materials and Methods

An exhaustive and database-assisted literature search was conducted to compile the results of multidisciplinary studies on the predictive modelling of soil nutrient dynamics using statistical and machine learning techniques for the effective management of soil nutrients. The literature search was executed within the Scopus

database to retrieve high-quality, peer-reviewed and indexed scientific literature on the topic. An extensive set of key terms was used to design the literature search strategy, including terms on soil fertility, N, P, K, organic carbon, spatial modelling, time series, machine learning, artificial intelligence, remote sensing and sustainable soil management. The results of the literature search yielded 4556 author keywords and to remove the effect of noise in the results, the minimum number of keyword occurrences was set to five to obtain the most promising results on the topic. Out of the total number of author keywords, 365 met the conditions to obtain the results on the topic.

The methodological framework of this study, spanning the period 2000 to 2025, comprised three complementary bibliometric analyses: i) co-occurrence network mapping, to identify dominant research clusters and conceptual interrelations among keywords, conducted using VOSviewer software, ii) thematic mapping and research theme classification, to categorise themes into motor, basic, niche and emerging/declining themes based on their centrality and density; and iii) annual scientific production and temporal trend analysis, conducted using the R programming language, to examine the year-wise distribution of publications and identify periods of growth in the field.

Bibliometric mapping and research trend analysis

Co-occurrence network mapping of research themes

Keyword co-occurrence analysis was performed using VOSviewer (version 1.6.20, van Eck and Waltman), applying the association strength normalisation method with a minimum keyword occurrence threshold of 5. Clustering was carried out using VOSviewer's built-in modularity-based clustering algorithm, with the resolution parameter set to 1.0 (default). The resulting network (Fig. 1) revealed prominent research clusters centered on soil fertility, organic carbon, remote sensing, RF, machine learning, artificial intelligence and nutrient variables such as nitrogen and potassium. Node size represents keyword occurrence frequency, while link thickness reflects the strength of co-occurrence association between keyword pairs. Node colour denotes cluster membership, indicating distinct but interconnected thematic groupings within the research domain.

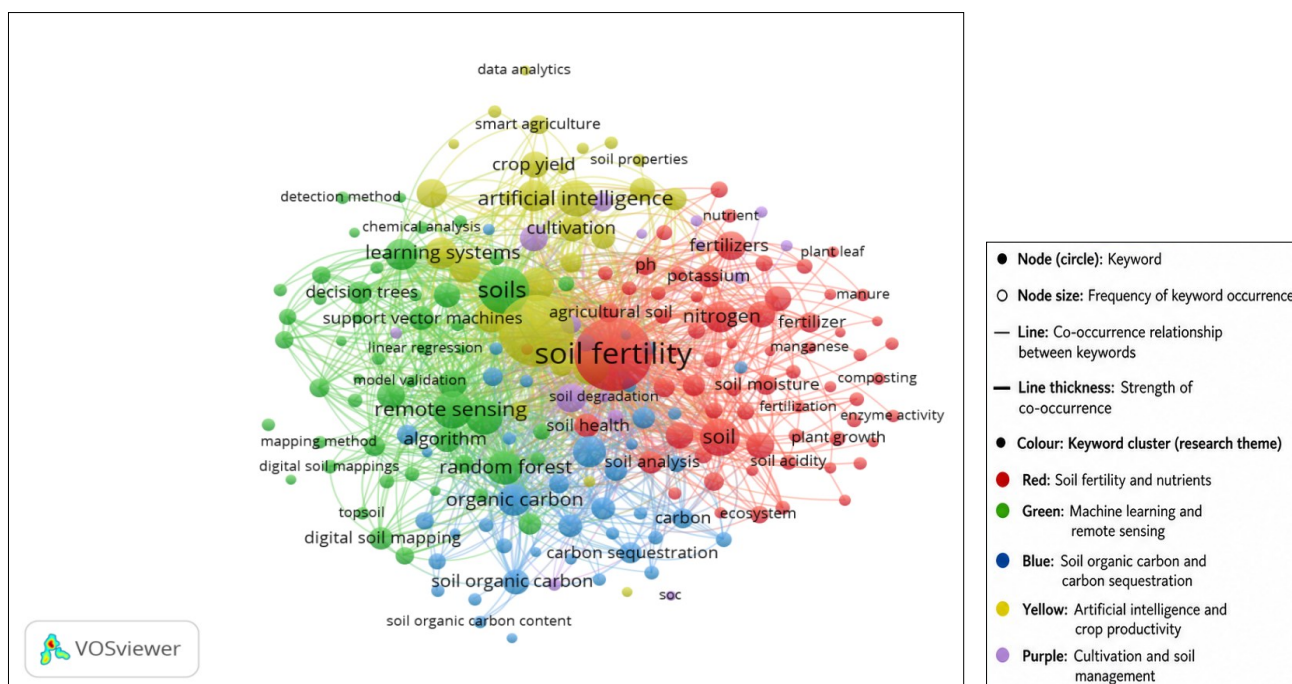


Fig. 1. Keyword co-occurrence network visualisation of research themes.

Thematic mapping and research theme classification

Thematic mapping was performed using the bibliometrix package in R, applying the thematicMap function based on Callon's centrality and density measures, with Louvain clustering and a minimum keyword frequency threshold of 5. The conceptual structure of the research field based on keyword centrality (x-axis) and density (y-axis). The resulting strategic diagram (Fig. 2) classified themes into 4 quadrants such as motor, basic, niche and declining/emerging based on relevance (centrality) and development (density) degree. Soil fertility and soil systems emerged as motor themes with high relevance and development, while remote sensing, soil organic matter and RF formed well-connected basic themes. Machine learning and artificial intelligence appeared as declining/emerging themes, indicating evolving methodological directions in soil predictive modelling.

Annual scientific production and temporal trend analysis

Annual scientific production was analysed using the bibliometrix package in R, with publication counts extracted from the bibliographic dataset and plotted against the corresponding year of publication (Fig. 3). The analysis covered publications from 2000 to 2025, with annual growth rate calculated to quantify the temporal trend in research output. Results indicate a relatively stable and low publication rate from the early 2000s through 2017, followed by a marked exponential increase from 2018 onward. This surge reflects the growing global research interest in data-driven soil nutrient

prediction and sustainable soil management strategies. The steep rise post-2018 aligns with the increasing adoption of machine learning and remote sensing technologies in agricultural research.

Overall methodological framework of the study

The total methodological framework is based on the use of literature search techniques, keyword filtering, co-occurrence networks, thematic mapping and production time series to provide an understanding of the evolution of research, methodological advancements and interdisciplinary integration in predictive soil nutrient modelling studies.

Fundamentals of soil nutrient dynamics

Soil nutrient components and their systematic transformations

Nutrient availability in soils is considered one of the basic determinants of plant growth, productivity and ecosystems (14) (Fig. 4). The chemical and physical forms and quantities of nutrients present in soils are constantly being modified by various processes in soils (15). Mechanisms such as mineral weathering, organic matter mineralisation and biological processes control and govern the transformations and bioavailability of nutrients (7). The relationship between soils, plants and the atmosphere plays a significant role in the uptake and cycling of nutrients. Macronutrients specifically nitrogen, phosphorus and potassium largely determine soil fertility and crop productivity (16). Environmental conditions, climate, topography and soil properties govern the movement and retention

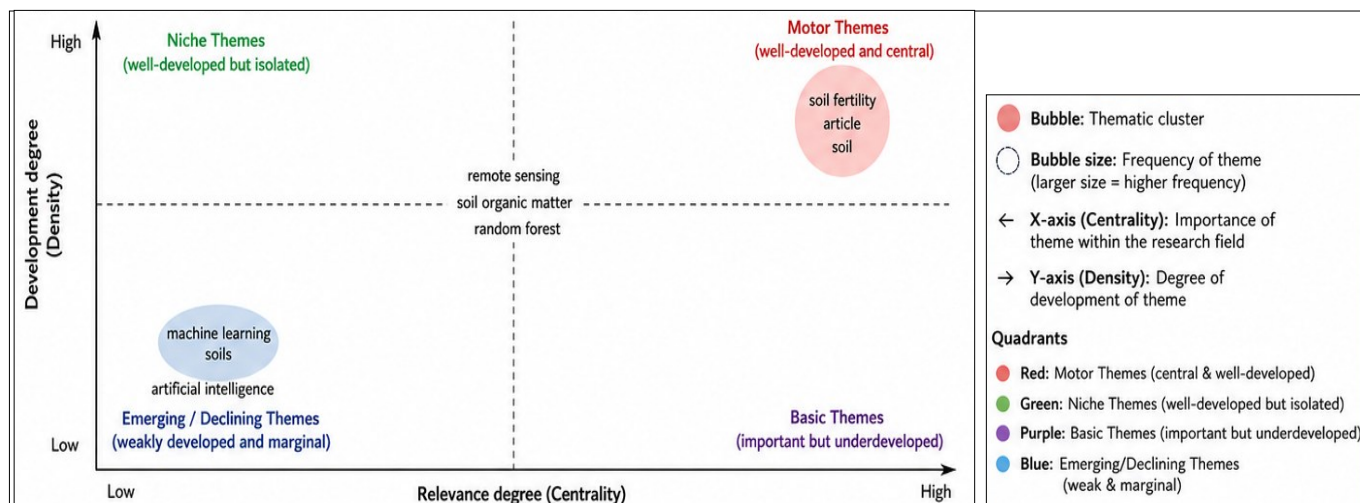


Fig. 2. Thematic mapping of research themes based on centrality and density.

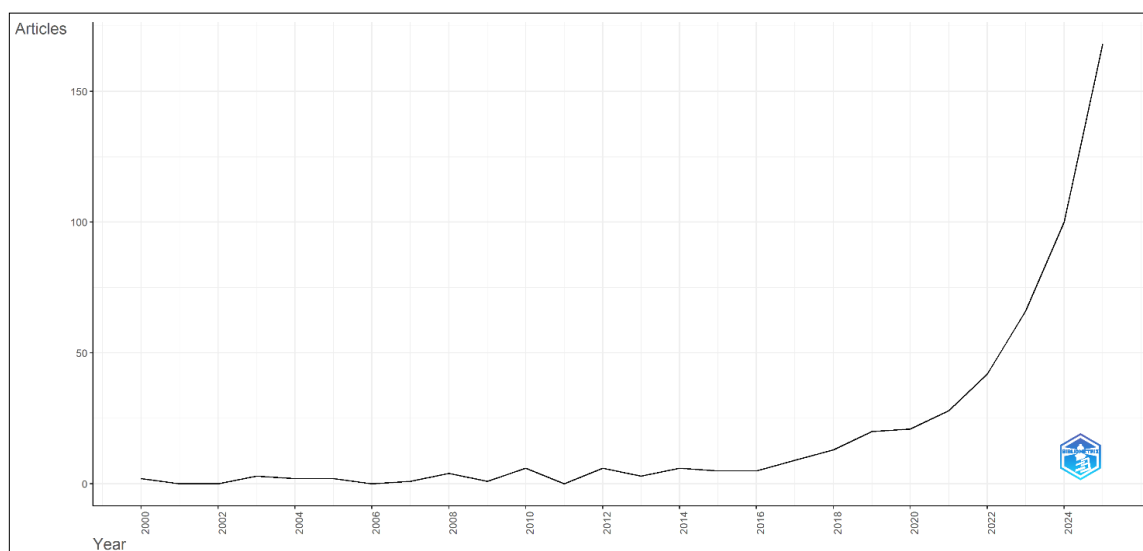


Fig. 3. Annual scientific production and publication trend analysis.

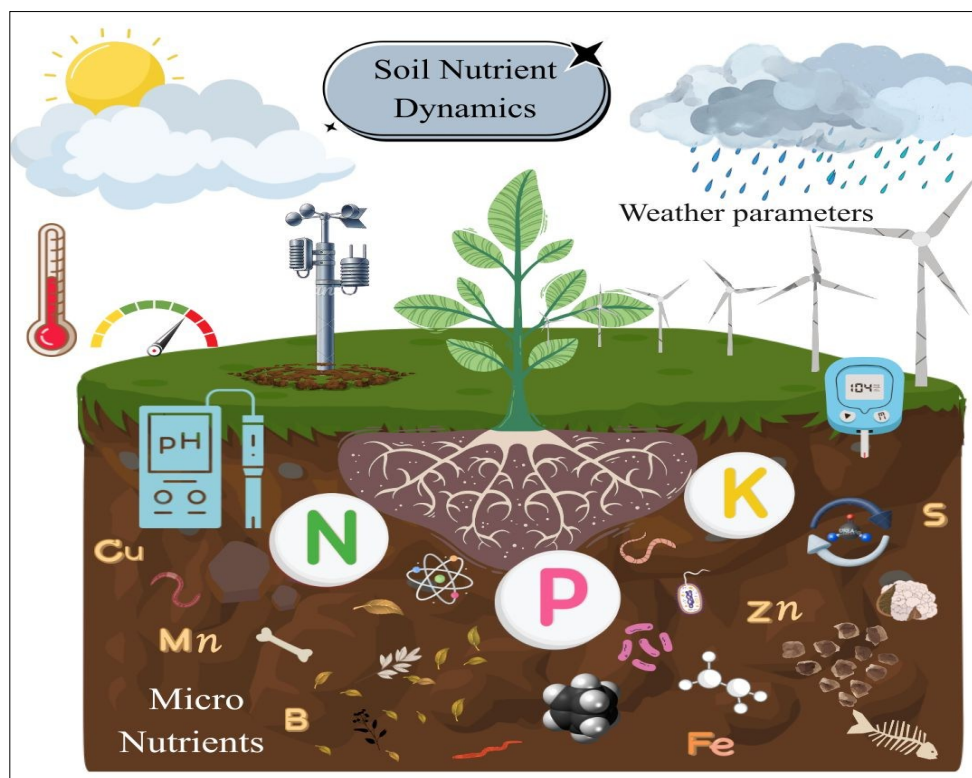


Fig. 4. Soil nutrient dynamics and predictive factors.

of nutrients (17). Nutrient cycling processes are inherently dynamic and complex. A rigorous and thorough understanding of these dynamic and complex nutrient cycling processes is considered basic and essential in the management and maintenance of soil fertility.

Factors influencing nutrient availability and transformation

Soil nutrient availability and transformation processes are controlled by several factors, including soil pH, organic matter, temperature, moisture status and texture, which include sand, silt and clay content (18). Climate, rainfall and temperature changes significantly affect nutrient availability in soil systems. Erosion causes nutrient depletion, including macronutrient and micronutrient availability, leading to decreased fertility with increased soil depths (19). The microbial population in soil regulates nutrient availability through biological processes that either increase or decrease nutrient availability to plants (20). A comprehensive understanding of these factors is essential to maintain soil nutrient balance, optimise fertiliser use and increase agricultural productivity (21).

Comparative influence of predictor variables on model accuracy

Predictor variables differ considerably in their contribution to soil nutrient prediction accuracy across studies. Soil physicochemical properties, such as pH, organic carbon and texture, remain the strongest and most direct predictors of nutrient availability (4). Topographic variables, including elevation and slope, improve accuracy mainly in spatially heterogeneous landscapes, while weather variables such as rainfall and temperature are more influential in time-series forecasting models (19). Management practices, including fertilisation history and irrigation, enhance prediction accuracy but are often underrepresented due to limited data availability (9). Remote sensing indices, such as normalised difference vegetation index (NDVI), support large-scale predictions with wide spatial coverage, whereas proximal sensing data, including hyperspectral and near-infrared sensors, offer higher-resolution, field-scale accuracy at greater cost (12). Therefore, models integrating multiple predictor categories, particularly soil properties combined

with remote or proximal sensing data, consistently outperform single-source models, highlighting the value of multi-source data integration in improving predictive accuracy.

Temporal and spatial variability in soil systems

Nutrients in the soil also show significant spatio-temporal changes, influenced by the changing environment, climate change, rainfall and temperature in various regions (19). Such changes have become vital to be monitored in the context of modern agricultural practices to meet the nutrient requirements and avoid nutrient deficiencies in the soil. Time series analysis and accurate forecasting models play an important role in understanding the long-term changes in soil fertility and nutrient availability in the soil. Such changes in the spatial variability of the soil in different depths and regions require nutrient management strategies to be implemented in the field, along with the application of fertilisers (21).

Given the complex, non-linear nature of these spatio-temporal patterns, machine learning models are increasingly used to capture such variability, which has driven growing interest in model explainability techniques to interpret their predictions. Methods such as shapley additive explanations (SHAP) and local interpretable model-agnostic explanations (LIME) help quantify the contribution of individual predictor variables, such as rainfall, temperature and soil depth, to specific nutrient predictions, thereby improving model transparency. Permutation importance further enables ranking of variables based on their overall influence on model accuracy (22), while partial dependence analysis illustrates how nutrient predictions change across the range of a given variable, holding others constant. Identifying such key drivers of soil nutrient variability has been highlighted as important for improving model interpretability in intensive agroecosystems (23). Incorporating these explainability techniques into spatio-temporal nutrient forecasting enhances trust in model outputs and supports more informed, site-specific nutrient management decisions.

Statistical approaches in soil nutrient prediction

Time series analysis techniques

Time series analysis evaluates soil nutrient parameters collected at regular intervals, allowing for the determination of significant patterns while avoiding random and seasonal changes (24). The autoregressive integrated moving average (ARIMA) model has proven its potential in forecasting various soil macronutrients like N, P and K, as well as micronutrients like iron, manganese, zinc and copper in various topographical regions (25). This model works by de-trending and de-seasonalising the parameters through difference, which is effective in the analysis of non-stationary parameters like soil nutrient content (21). This model has further been enhanced by the introduction of the seasonal autoregressive integrated moving average (SARIMA) model (Table 1), which is effective in the short-term forecasting of various parameters in agricultural regions (26). Probabilistic forecasting based on the historical records of the databases related to the nutrient content of the soil is effective in critical agro-based decision-making processes like the determination of the amount of fertiliser required, the choice of crops, the determination of the time required for irrigation and the optimisation of crop production (27).

Multivariate statistical methods

Advanced data reduction techniques play a vital role in analysing patterns of soil nutrients, using multivariate statistical tools (28), such as principal component analysis (PCA) (29), factor analysis, canonical correlation analysis (20) and discriminant analysis, which can be used to deal with data dimensionality and extract principal patterns

from multiple variables (Table 1). The PCA helps in differentiating agricultural fields based on various components, which account for maximum variance in agricultural soil data, incorporating chemical properties like effective cation exchange capacity (ECEC), cation exchange capacity (CEC), calcium, magnesium, organic carbon, total nitrogen and pH, along with physical properties like bulk density and porosity, which explain more than 55 % of total variation in soil productivity (30, 31). Factor analysis extends this by identifying basic factors present in observed variables, which helps in identifying clusters of nutrients and their covariance structures, using fewer latent variables; for instance, pH exhibited high consistency, with 89.5 % of its total variance being explained by its factors, which indicates its high association with general trends in soil chemistry (24). Such statistical tools generate uncorrelated linear combinations of original variables, which can be easily interpreted and can be used to isolate pivotal variables like N, P and K, which are affected by environmental and anthropogenic factors. The integration of correlation analysis, regression analysis, ANOVA and descriptive statistics with multivariate statistical methods can provide a comprehensive assessment of soil health, which can be used to forecast reactions in soils, recommend fertilisers and choose crops depending on different types of soils and fertiliser levels to improve crop yields and quality (32). The application of statistical methods can be used in soil classification, precision farming, defining site-specific management zones, analysing metal contamination, developing fertiliser application and analysing large data sets of physical, chemical and biological parameters of soils individually and collectively (33).

Table 1. Statistical and machine learning approaches for soil nutrient dynamics

Statistical approach / category	Specific models	Primary application	Key advantages and performance	Reference
Multivariate methods	PCA, Factor Analysis, Discriminant Analysis	Dimensionality reduction for pH, Carbon and ECEC.	Explains over 55 % of total soil productivity variation.	(24)
Classical machine learning	RF, SVM, PLSR, GAM	Predicting spatial distribution of nutrients and texture.	Consistently outperforms linear approaches (PLSR) in complex terrains.	(25)
Time series analysis	ARIMA, SARIMA, ARIMAX	Forecasting N, P, K and micronutrient levels over time.	Filters noise and seasonality in non-stationary soil data.	(26)
Spatial Statistical	Ordinary, Regression and Indicator Kriging	Mapping nutrient distribution at unsampled locations.	Minimizes estimation errors using spatial correlation.	(37)
Machine learning	PLSR, BPNN, GA-BPNN	Prediction of total potassium (TK) in soil	The GA-BPNN model achieved the highest prediction accuracy ($R^2 = 0.90$, $RRMSE = 20.37\%$) for total potassium (TK), outperforming both PLSR and BPNN by effectively capturing nonlinear relationships.	(42)
Machine learning	RT and RF	Prediction of soil texture	RF outperformed RT, providing higher prediction accuracy and better handling of complex nonlinear relationships.	(43)
Deep learning	CNN, LSTM, ANN, GRU	Real-time sequence modelling for nutrients and crop suitability.	Captures complex non-linearities with > 90 % accuracy in large datasets.	(53)
Ensemble learning	XGBoost, AdaBoost, Stacking, Bagging	Predicting pH, EC and available potassium via model integration.	Achieves > 90 % accuracy; reduces RMSE by up to 72 %.	(63)
Hybrid model frameworks	ARIMAX-ANN, ARIMA-LSTM, XGBoost-LSTM	Forecasting nitrogen and nutrient consumption patterns.	Merges linear trends with non-linear residual modelling for reliability.	(67)

AdaBoost: adaptive boosting; ANN: artificial neural network; ARIMA: autoregressive integrated moving average; ARIMAX: autoregressive integrated moving average with exogenous variables; BPNN: backpropagation neural network; CNN: convolutional neural network; GA-BPNN: genetic algorithm-backpropagation neural network; GAM: generalised additive model; GRU: gated recurrent unit; LSTM: long short-term memory; PCA: principal component analysis; PLSR: partial least squares regression; RF: random forest; RT: regression tree; SARIMA: seasonal autoregressive integrated moving average; SVM: support vector machine; XGBoost: extreme gradient boosting.

Spatial statistical models

Geostatistical techniques like kriging have become an integral part of estimating soil properties at unsampled locations by accounting for their spatial autocorrelation (34) and minimising estimation errors (35). The semivariogram is the fundamental component of geostatistical analysis that offers significant insights into understanding soil properties in terms of their spatial variability (36). Various geostatistical techniques like ordinary kriging (OK) (37), regression kriging (RK) (38) and indicator kriging are employed based on the complexity of soil properties. The geostatistical techniques generally use spherical, exponential and Gaussian models to explain the spatial dependency of soil properties, which varies from moderate to strong (39). The geostatistical analysis offers significant advantages in terms of assessing uncertainties with the help of conditional cumulative distribution functions, which enables the prediction of threshold exceedance probabilities (35). The integration of auxiliary environmental variables with geostatistical techniques like kriging has been found to improve prediction accuracy (40). The application of geostatistical techniques like kriging has been found to be extremely useful in soil nutrient management (39). In Table 1 a comparative summary of statistical and machine learning approaches used for soil nutrient dynamics prediction. It highlights their applications, performance, advantages and limitations for supporting sustainable soil fertility management.

Model generalisation across climatic zones, soil types and management systems

Statistical and geostatistical models often show reduced accuracy when applied beyond the climatic zones, soil types or management systems for which they were calibrated (41). Transfer learning and domain adaptation approaches, which fine-tune models using smaller target-region datasets or adjust for shifts in feature distributions, offer promising strategies to improve generalisation without complete retraining (38). Standardised calibration methods, including harmonised sampling and measurement protocols, further support consistent model performance across regions and laboratories (9). Thus, these strategies would enhance model robustness and support reliable application across diverse agro-ecological conditions.

Machine learning and advanced statistical models

Classical machine learning algorithms

Regression trees (RTs) and their ensemble extension, RF, are statistical models used to predict the spatial distribution of soil properties based on terrain parameters (42). The RTs use decision rules for partitioning the data set into homogeneous groups by implementing tenfold cross-validation for avoiding overfitting. The RF is an extension of the RT algorithm, where many unpruned trees are grown from bootstrap samples of the data set with random selection of predictor variables for each tree (22). The RF is based on the ability of the statistical model to handle categorical and continuous data without assuming any data relationships. The RF is found to outperform the RT models in predicting the components of the soil texture based on the importance measures of the RF algorithm. The RF can be viewed as a black box compared with the RT algorithm, but it provides better prediction accuracy than the RT models by avoiding overfitting through out-of-bag error estimates, making it a powerful tool for soil-landscape modelling applications (43). The RF is an ensemble learning method used for estimating the soil properties by building many regression trees based on the bootstrap samples of the data set. The process iteratively fits the

individual trees using a random selection of predictors (mtry) until the minimum node size is reached, while the out-of-bag error estimated from the test data converges as the number of trees grows. The RF maintains the stability of the model through low bias and variance, as it addresses the issue of over-fitting the individual decision trees. Research comparing the performance of RF, generalised additive model (GAM) and support vector regression (SVR) on the prediction of soil nutrient content indicates the superiority of RF, as it presents better performance through lower Root mean square error (RMSE) and less variability. The RF method for soil nutrient prediction is highly recommended due to its robust predictive capacity, fewer parameter tuning steps, simplicity of the model building process and higher interpretability of the results compared to other machine learning techniques (44). The use of machine learning algorithms for the prediction of soil nutrient dynamics has revolutionised the field and support vector machine (SVM) presents good performance as a robust tool when coupled with visible and near-infrared (Vis-NIR) spectroscopy. The SVM model presents better performance while handling large datasets from diverse geo-pedological conditions. Through the application of automated grid search optimisation and repeated 10-fold cross-validation, the method exhibited high levels of predictive accuracy for total nitrogen (TN) and organic carbon (OC), with R^2 values ranging from 0.82 to 0.91 being obtained for the top and subsoil strata. This method outperforms traditional PLSR, owing to its ability to identify intricate patterns within high-dimensional spectral data sets, especially in soils with higher organic matter content (45). Empirical evidence has consistently shown that the most efficient methodology for estimating various soil properties such as OC, TN, clay content and pH using Vis-NIR spectroscopy is the SVM method (Table 2). This sophisticated method of prediction is not only non-invasive and cost-effective but also highly accurate in its application for monitoring nutrient variability within soils of varying depths and under different conditions (46). The RF is another algorithm that uses decision trees and combines their predictions to produce more accurate results. This method uses sampling and randomisation of features to avoid overtraining on the data set. The SVM model is based on the fundamental principle of maximising the margin between the support vectors and the decision hyperplane, using kernels to map low-dimensional input features into a high-dimensional space for optimal performance of the classifier (47). During the comparative study of the performance of these machine learning models on different types of agricultural land use and climatic conditions, it was found that the RF algorithm was more stable and accurate in predicting the stoichiometric relationship of C : N : P, while the SVM algorithm performed better in predicting parameters such as soils in vegetable fields and net N mineralisation rates in rice paddies. The relative stability of the models also varied according to the type of land use; for example, RF > XGB > GBDT > SVM for vegetable fields and SVM > XGB > RF > GBDT for rice paddies in predicting net N mineralisation rates (23). The workflow of the proposed predictive modelling framework, including data collection, preprocessing, statistical modelling, model validation and nutrient prediction are illustrated in Fig. 5. The framework supports soil fertility assessment and sustainable nutrient management through accurate predictive analysis.

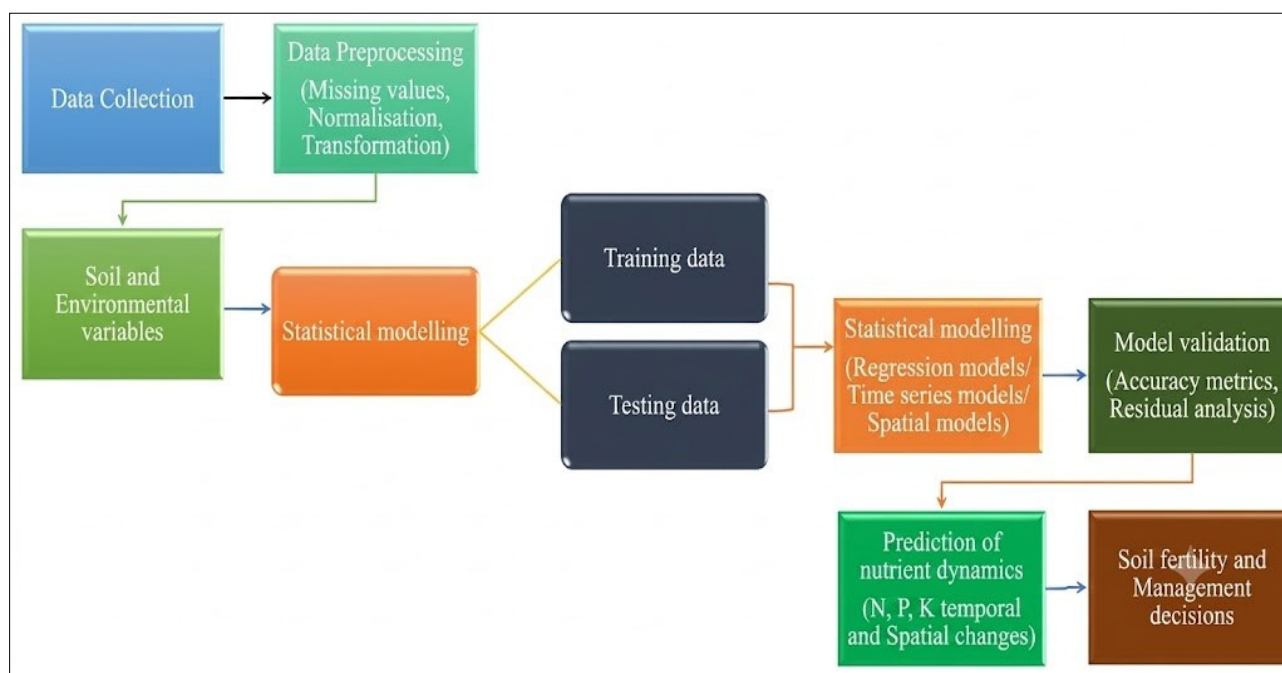
Emerging technologies and future modelling directions

Beyond classical machine learning, emerging approaches such as transformer architectures and graph neural networks offer potential for capturing complex temporal and spatial dependencies in soil

Table 2. Comparative performance and limitations of soil nutrient prediction models

Models	Dataset	Models performance measures	Limitations	Reference
PLSR, SVM, RF (NIR-based)	89 orchard samples (Guangzhou, China).	RF best: N and P ($R^2=0.54$), K ($R^2=0.43$).	Weak K spectral response; preprocessing sensitive.	(12)
RF and SVR (Micronutrients)	68 topsoil samples (NE Iran).	RF best: NRMSE 91 % (Fe), 94 % (Mn), 91 % (Zn).	Covariate sensitivity; SVR missed extremes.	(13)
ML vs. DL (CNN, LSTM)	Kaggle public dataset (N, P, K, Climate).	DL models best: >90 % accuracy; CNN/LSTM superior.	ML weak for non-linear; DL needs big data.	(19)
RF, SVM, XGBoost, GBDT	210 samples (Eastern China).	XGBoost best: Avg $R^2 > 0.81$, RPD >2.6.	Variation by land-use and climate.	(23)
RF and NN (N, P, K prediction)	Lab-synthetic and soil samples (Agra, India).	RF best: P_2O_5 (23.6 % error), K_2O (16 % error).	Synthetic data; limited generalisation.	(42)
RT and RF (Soil texture)	56 profiles (Ecuadorian Andes).	RF outperformed RT.	High uncertainty; small dataset.	(43)
GAM, RF, SVR (C, N, P)	139 topsoil samples (South Korea).	RF best: RMSE (C: 17.5, N: 1.13 mg/g); P correlation ($r=0.75$).	GAM uncertainty; SVR tuning.	(44)
PLSR, BPNN, GA-BPNN	75 soil samples (Guangdong, China).	GA-BPNN best: $R^2=0.90$, RRMSE=20.37 % for TK.	Low linear accuracy; BPNN tuning issues.	(45)
PLSR and SVM (Vis-NIR)	105 samples (Mine-affected, Iran).	SVM best: TN ($R^2=0.91$), OC ($R^2=0.87$).	PLSR weak in polluted soils; depth sensitive.	(46)
SVM and Decision Tree	1700 samples (Chhattisgarh, India).	SVM (kernel) best: 92 % accuracy; F1-score 91 %.	Poor linear performance; data balance issues.	(47)
Ensemble (SVM, ANN, RF, XGB)	204 samples (Andean Peru).	Weighted ensemble best: $R^2 > 0.93$; 72 % RMSE reduction.	Local EC variability; ANN overfitting.	(63)
PLSR, SVR, AdaBoost	188 samples (Anhui, China).	AdaBoost best: $R^2 = 0.94$, RMSE = 31 mg/kg.	Linear/SVR low stability; preprocessing sensitive.	(64)
ARIMA, ML, Hybrid	India time-series (1960–2023).	XGB/RF best for consumption; ARIMA for trade.	Hybrid gains negligible for stable series.	(67)

AdaBoost: adaptive boosting; ANN: artificial neural network; ARIMA: autoregressive integrated moving average; ARIMAX: autoregressive integrated moving average with exogenous variables; BPNN: backpropagation neural network; CNN: convolutional neural network; DL: deep learning; GA-BPNN: genetic algorithm–backpropagation neural network; GAM: generalized additive model; GRU: gated recurrent unit; LSTM: long short-term memory; ML: machine learning; NN: neural network; PCA: principal component analysis; PLSR: partial least squares regression; RF: random forest; RT: regression tree; SARIMA: seasonal autoregressive integrated moving average; SVM: support vector machine; SVR: support vector regression; Vis-NIR: visible and near-infrared; XGBoost: extreme gradient boosting.

**Fig. 5.** Methodological framework for predictive modelling of soil nutrient dynamics.

nutrient data, building on existing deep learning frameworks (48, 49). Physics-informed and federated learning approaches may further improve model reliability and enable collaborative training across farms without sharing raw data. Edge artificial intelligence (AI), supported by advances in electronic and proximal sensing (50), enables real-time, on-site nutrient prediction, while UAV and satellite-based sensing continue to expand high-resolution spatial data availability for large-scale monitoring (51). Foundation models pre-trained on diverse agricultural datasets also hold promise for enabling transfer learning across regions and crop systems, representing a key forward-looking direction for the field.

Deep learning neural networks

The application of deep learning neural networks (52) in soil nutrient forecasting has been identified as an innovative technique in precision agriculture that utilises sophisticated neural networks such as convolutional neural networks (CNN) (53), long short-term memory (LSTM) (54), ANN (55) and gated recurrent units (GRU) (48) to examine complex datasets that include soil parameters, pH values and nutrient content (Table 2). These architectures differ considerably in their suitability, computational demands and interpretability, making architecture selection dependent on the nature of the available data and the specific prediction task. The ANN models are preferable for simpler, static, tabular soil datasets with limited feature interactions, offering low computational cost and moderate interpretability, but they are less effective in capturing temporal or spatial dependencies. The CNN models are best suited for spatial or image-based data, such as remote sensing imagery and hyperspectral soil scans and require moderate-to-high computational resources along with large, labelled datasets, though their internal feature representations remain relatively difficult to interpret. The LSTM and GRU models are specifically designed for sequential, time-series data, such as seasonal nutrient variation, rainfall and temperature patterns and are highly effective in capturing long-term temporal dependencies; however, they demand substantial computational resources and large historical datasets and their internal mechanisms offer limited interpretability compared to simpler models. Notably, GRU architectures require fewer parameters and lower training time than LSTM while achieving comparable accuracy, making them a computationally efficient alternative for resource-constrained applications. Deep learning models have shown greater predictive accuracy of over 90 %, thus proving superior to other machine learning techniques in precision agriculture (56). The ability of deep learning models to identify long-term relationships in complex datasets by sequentially processing nitrogen (N), phosphorus (P), potassium (K), soil pH values, humidity, temperature and rainfall parameters has been highly beneficial in precision agriculture. The CNN model (57) has been identified as highly effective in image classification in precision agriculture to determine nutrient content in plants, whereas the LSTM model has been identified as highly effective in sequence prediction to determine soil-plant interactions in diverse environmental conditions. The implementation of these models typically follows a three-phase process: (i) collection of datasets with soil and environmental parameters, (ii) feature extraction using deep learning algorithms (49) appropriate to the data structure and (iii) crop recommendation based on nutrient prediction and soil fertility indices. Overall, the benefits of deep learning models (58) include accurate real-time prediction of crop suitability and fertilisation needs, cost-effectiveness through automation, adaptability to local conditions and early warning capability for nutrient depletion in

intensive farming systems, provided the architecture is matched appropriately to data type and available computational infrastructure (19).

Ensemble learning techniques

Ensemble learning combines models to increase prediction accuracy using two main techniques: boosting and bagging. Boosting techniques, including AdaBoost, XGBoost (59), light gradient boosting machine (LightGBM), categorical boosting (CatBoost) and histogram-based gradient boosting (HistGradientBoosting), work by converting weak models to strong models by minimising loss functions using gradient descent optimisation methods. The RF, which is one of the most popular ensemble methods for bagging, consists of several classifiers trained in a random manner. Ensemble methods, including XGBoost and LightGBM, have been reported to work effectively in soil classification problems, with accuracy rates higher than 90 %. These methods, including tree-based ensemble methods, especially gradient-boosting methods, have been observed to work effectively even with small amounts of training data. These ensemble methods have been observed to give better results than conventional machine learning methods, which make them important in solving predictive modelling problems in soil analysis (60). The predictive modelling of soil nutrient dynamics has been significantly improved using ensemble machine learning methods, including SVM, ANN (61), RF and XGBoost (62) using simple averaging and stacking methods (Table 2). The process of implementation incorporates two types of ensemble methods: (i) the average ensemble method, which strictly adheres to the principle of equal weighting in the "wisdom of crowds" concept and (ii) the performance-weighted ensemble method, which incorporates the stacking optimisation principle based on spatial cross-validation scores, assigning higher weights to models with superior performance. In the experiments conducted, the ensemble models have consistently outperformed the individual models by effectively utilising the relative strengths of the models and avoiding the corresponding weaknesses of the models used in the experiment. The ensemble method has resulted in the achievement of R^2 values greater than 0.93, with significant improvements in the prediction accuracy of the models, where the RMSE and MAE have shown improvements of 72 % and 75 %, respectively, compared to the best models in the prediction of soil pH and electrical conductivity. The weighted ensemble method has shown remarkable effectiveness in the prediction of complex relationships by effectively balancing the predictions of the models used in the ensemble method, ensuring the maintenance of the spatial characteristics of the models through the rigorous cross-validation process (63). The AdaBoost and gradient boosting regression tree (GBRT) models have shown the best accuracy and stability in the experiment, with the testing R^2 values of the models showing values greater than 0.93, with the residual prediction deviation (RPD) values classified as A. The findings of this study reveal that incorporating appropriate spectral pretreatment techniques with ensemble learning models produces a consistent and robust framework to forecast soil-available K (64). The robust ensemble learning models significantly improve predictive modelling in precision agriculture and digital soil mapping in heterogeneous agroecosystems with high variability.

Hybrid model framework

The hybrid modelling approach was successful in improving the accuracy of soil nitrogen content prediction by combining linear and nonlinear modelling techniques. The hybrid modelling approach

divides time-series data into linear and nonlinear parts, which are then predicted using autoregressive integrated moving average with exogenous variables (ARIMAX) and ANN, respectively. The inclusion of external variables like OC in the ARIMAX model improves its performance significantly over traditional ARIMA models. The application of ML techniques was successful in demonstrating their potential in identifying and modelling the complex variations present in soil nutrient data. The hybrid modelling approach combining ARIMAX and ANN was successful in improving the accuracy of soil nutrient prediction compared to individual linear and nonlinear models. The cohesive modelling approach was successful in generating more accurate forecasts and the results indicated that 100 % NPK + FYM was the treatment combination that produced maximum levels of N in the soil (25). The hybrid modelling approach combined linear models like ARIMA and XGBoost with nonlinear models like LSTM and GRU (65). This framework begins with the detection of linear trends and short-term patterns through ARIMA or XGBoost, with the remaining residuals being predicted by the LSTM model to detect complex nonlinear patterns (66). These individual predictions are then combined additively to produce the final predictions for nutrient consumption patterns. However, it has been observed in various empirical studies that the performance enhancement of the model is negligible, especially in forecasting P and K levels, as hybrid models provide little benefit in comparison to standalone approaches. This is because the primary use of hybrid models is in the presence of low-frequency nonlinear residuals, as opposed to the presence of white noise residuals, which are generally negligible in the context of agricultural forecasting, as standalone models like RF and XGBoost are more practically useful (67). In Table 2 compares the performance of different soil nutrient prediction models based on their prediction accuracy, strengths and limitations reported in previous studies. It provides a comprehensive evaluation of statistical and machine learning models, highlighting their suitability for improving soil nutrient prediction and supporting sustainable soil fertility management.

Challenges and limitations

A major challenge here is the accurate representation of the large spatial and temporal variations of soil properties and nutrients. Even using dense sampling strategies and sophisticated geostatistical techniques, it has been found that a large proportion of the variability remains unexplained, especially for phosphorus and potassium, which exhibit large coefficients of variation and only moderate predictive performance (68). Similarly, linear and mixed effects models can only account for a small proportion of the variance of the concentrations or uptake of nutrients and the accuracy of such predictions can only be achieved for a limited number of nutrients such as K or boron, while the remaining macro- and micronutrients can only be predicted inaccurately (69). Similarly, national-scale models for biological indicators such as bacterial richness can only achieve adjusted R^2 values of 0.56, failing to account for any temporal variations such as seasonal changes (70). The accuracy of the models can also be limited due to the use of limited or uncertain covariates such as missing information on management practices such as the timing of fertiliser application, crop variety, or drought occurrence, as well as the use of coarse scale climate information, which is known to affect nutrient dynamics but for which such information may not be readily available (51). The statistical models show sensitivity to multicollinearity in soil variables, which can affect model accuracy, as well as to errors in

predictors, which can affect predictions and, by extension, estimated trends in soil OC or nutrient status, particularly when sampling design and data are not rigorously controlled (71). Furthermore, classical linear regression analysis may not fully capture nonlinear responses to soil nutrients, moisture, or contaminants, with several studies comparing accuracy using linear regression analysis with machine learning techniques, which show that, for instance, using RFs, it is more accurate to model the nonlinear response of soil nutrient elements on plant uptake, thereby showing another limitation of using classical linear statistical analysis (72). Although these models provide accuracy metrics, the uncertainty is rarely spatially explicit, hampering the reliable application of predictions for decision-making in precision fertilisation (50).

Training data often have characteristics of clustering and non-representativeness, which restrict the transferability of models and the accuracy of maps on a large scale (73). The lack of detailed or fine-resolution data on parent material, geology, land use and management practices constrains the ability of models to identify and represent the major drivers of nutrient variation (41). The spectral data from Vis-NIR, hyperspectral and X-ray fluorescence (XRF) measurements often show poor or indirect relationships with certain nutrients and the presence of high matrix effects further complicates the development of models (74). No single algorithm consistently outperforms all others and the use of simple linear or partial least squares regression (PLSR) models may not capture nonlinear relationships, while the use of complex machine learning/deep learning models can result in over-fitting, high computational costs and interpretability issues (75). The class-based prediction of fertility or nutrient status is often prone to errors due to misclassifications of boundaries (76). A considerable number of studies focus only on a few macronutrients, whereas micronutrients and multi-nutrient interactions, which play an important role in yield responses, are often neglected (77). In addition, the use of hyperspectral sensors, handling of high-dimensional data sets and application of sophisticated machine learning workflows require specific skills and facilities, which limit their application in routine soil testing (78).

Data, modelling and computational constraints

Reliable nutrient prediction depends on minimum, property-specific sampling densities rather than simply larger datasets; work in the Mayenne region and at national scale showed accuracy gains plateauing beyond a threshold density (~ 1 sample/2 km²), with tiered classes proposed to grade map reliability (79). A related constraint is the lack of harmonized sampling, preparation and analytical protocols; initiatives like the Food and Agriculture Organization (FAO) Global Soil Laboratory Network (GLOSOLAN) address this through standard operating procedures and inter-laboratory proficiency testing, though wider adoption including standardised metadata on depth, timing and sensor calibration is still needed (80). Model transferability remains limited, as tree-based algorithms are constrained to the covariate range of their training data and often perform poorly outside it; transfer learning, domain adaptation and similarity indices (e.g., multivariate environmental similarity surfaces) offer partial solutions by flagging unreliable extrapolation zones (81). Uncertainty is also rarely quantified spatially: quantile regression forests and conformal prediction provide calibrated, non-parametric prediction intervals that outperform classical variance estimates and closer adherence to standards such as GlobalSoilMap's 90 % prediction-interval

requirement would improve decision-relevance (82). Finally, model choice involves a computational interpretability trade-off: tree ensembles (RF and XGBoost) balance moderate cost with reasonable interpretability via SHAP, while deep learning and GAM architectures demand greater infrastructure and offer less transparency, limiting adoption where computational resources or machine learning expertise are constrained (83).

Farmer usability and decision-support applications

Translating predictive nutrient models into on-farm benefit requires outputs in an actionable form. Geographic information system (GIS)-based decision-support tools integrating digital soil maps have enabled variable-rate fertilisation prescriptions, letting farmers target inputs spatially and reducing waste, while coupling with unmanned aerial vehicle (UAV) multispectral/hyperspectral sensing supports near-real-time, in-season updates (84). However, adoption hinges on affordability, reliability and interpretability as much as accuracy: survey evidence from European farmers shows uncertainty about yield/profit impact and acquisition/maintenance costs remain key adoption barriers (85). Closing this gap requires presenting predictions alongside clear spatial uncertainty, simplifying outputs into prescription maps compatible with existing machinery and providing advisory support so smaller or less-resourced farms can also benefit, not only well-capitalised operations.

Conclusion

The predictive modelling of soil nutrient dynamics has been recognised as a reliable and vital tool for effective and sustainable management of soil nutrients. The review established that statistical models, such as time series models, multivariate models and geostatistical models, form a strong foundation for understanding the spatio-temporal dynamics of soil nutrients. The use of machine learning models, ensemble models and deep learning models appreciably improves the accuracy of the predictions. Among the models, the ensemble models and tree-based models have been identified as superior models to the conventional linear models, particularly when modelling heterogeneous agro-ecosystems. However, some challenges that are critical to the models include issues of data quality, spatial representation, multicollinearity and covariate limitations. Therefore, the use of quality data and effective statistical and machine learning models is essential for improving the accuracy and applicability of the models.

Future research should prioritise the development of standardised benchmark datasets, open-source model repositories and integration of multi-source sensing data (satellite, proximal sensors, unmanned aerial vehicle (UAVs)) to improve model comparability and spatial-temporal resolution. Incorporating uncertainty quantification and explainable artificial intelligence (XAI) will be essential for improving prediction reliability and model transparency, while extensive field-scale validation across diverse agroecological conditions is needed to confirm real-world robustness and applicability. The advancements in predictive modelling can ensure accurate fertiliser recommendations, optimal nutrient use efficiency and environmentally friendly agricultural practices, which are in line with sustainable soil nutrient management.

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Authors' contributions

RR prepared and drafted the manuscript. BS provided guidance and supervision throughout the writing process. RGS, JP and KGS critically reviewed and revised the manuscript. All authors read and approved the final version of the manuscript.

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